

SCOPE : L7–Insight

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Chapter 0: Introduction

The Sephirotic names used in this paper (Malkuth, Yesod, Keter, and others) are not used in a religious or mystical sense. Within SCOPE, they are borrowed as labels to identify cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for estimating the cognitive structure of humans and AI. Here, "coordinate system" does not mean a flowchart showing cognitive processing as a fixed sequence. Each layer does not represent a hierarchy or a temporal order.

Within SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as "Malkuth." What can be observed is the output itself; it is not possible to directly know what cognitive structure operated internally. For this reason, SCOPE uses observable output as a clue to infer the cognitive structure behind it.

Expressions such as "Make it feel a bit more blue" or "Something's off" may not be fully understood from the words alone. At the same time, among those who share experience in the same field, intent can be shared through these short expressions alone. These are not simply a lack of information; they may represent a large amount of experience expressed in a highly abstract, highly compressed form.

This paper addresses one such cognitive structure: L7. L7 is a cognitive layer responsible for highly abstract, highly compressed cognition formed through the accumulation of experience. Its definition and characteristics are discussed in detail in Chapter 1.

Note that L7 is not defined as the starting point of cognitive processing. SCOPE does not assume that cognition begins from any particular layer; each layer is treated as part of a cognitive structure in which layers mutually influence one another.

This paper examines the characteristics and basic properties of L7, as well as its possible functional correspondence in humans and AI, centered on its role as a cognitive layer of highly abstract, highly compressed cognition derived from experience. Some points remain at the hypothesis stage and are left as topics for future consideration.

Chapter 1: What Is L7

L7 is a cognitive layer responsible for highly abstract, highly compressed cognition formed through the accumulation of experience.

Here, "experience" does not mean a mere quantity of knowledge. It includes the sensations, insights, and highly abstract concepts formed through engagement with, repetition of, and observation of a given subject. Within SCOPE, L7 is positioned as the cognitive layer that organizes and estimates this experience-derived cognition.

L7 is not a cognitive layer that constructs explanatory models or theories. Nor is it a cognitive layer that guarantees the quantity of knowledge or logical correctness. It is positioned as the cognitive layer that organizes and estimates cognition formed from experience.

L7, as used here, does not refer to any specific biological organ or region of the brain. Within SCOPE, it is positioned as a cognitive layer for estimating what kind of experience-derived cognition has formed within a given individual.

This paper positions L7 as a cognitive layer of highly abstract, highly compressed cognition derived from experience, and organizes its characteristics in the chapters that follow.

Chapter 2: Basic Characteristics of L7

Chapter 1 defined L7 as a cognitive layer responsible for highly abstract, highly compressed cognition formed through the accumulation of experience. Building on that definition, this chapter organizes the basic characteristics of L7.

L7 is a cognitive layer responsible for cognition formed through the accumulation of experience. Here, "experience" does not mean a mere quantity of knowledge, but cognition formed through engagement with, repetition of, and observation of a given subject. However, this experience is not necessarily retained as knowledge; it may instead be formed as highly abstract, highly compressed cognition.

The cognition handled by L7 is not uniform. It includes a variety of forms, such as sensations, insights, and highly abstract concepts. These differ from one another in nature and in degree of abstraction.

L7 is also not a cognitive layer that retains experience as knowledge in its raw form. Cognition formed through the accumulation of experience may be retained in a highly abstract, highly compressed form. As a result, it may be expressed as short phrases, and the content understood from the same expression may differ depending on differences in experience.

Furthermore, cognition formed by L7 is not always fixed. Different sensations, insights, or highly abstract concepts may form depending on new experience or on consistency with the cognitive structure as a whole. However, the mechanism of this change requires

consideration that includes interaction with other layers, and is therefore not addressed in this paper.

This paper positions L7 as a cognitive layer with these basic characteristics. The next chapter organizes the conditions under which L7 may change.

Chapter 3: L7 Changes

Chapter 2 organized L7 as a cognitive layer responsible for highly abstract, highly compressed cognition formed through the accumulation of experience, and noted that it includes a variety of forms of cognition. This chapter examines the conditions under which L7 may change.

L7 is a cognitive layer whose basic role is to hold cognition formed through the accumulation of experience. New experience may lead to the formation of sensations, insights, or highly abstract concepts different from those held previously. Here, "change" does not mean the loss of experience, but the formation of different cognition.

Change in experience-derived cognition is not necessarily proportional to the amount of experience. This may also depend on the nature of the cognition being formed, but this paper does not go further into that point.

The process through which experience-derived cognition changes is not addressed in this paper. This change requires consideration that includes its relationship with other layers.

As described above, L7 is positioned as a cognitive layer that, while fundamentally forming cognition derived from experience, may form different sensations, insights, or highly abstract concepts depending on circumstances. The next chapter organizes how L7 may relate to other layers.

Chapter 4: L7's Interaction with Other Layers

Chapter 3 organized L7 as cognition formed through the accumulation of experience, and as a cognitive layer capable of change through new experience. This chapter examines how L7 may relate to other layers.

L7 is not a cognitive layer that functions in isolation. Cognition formed through the accumulation of experience is formed within interaction with other layers, and may in turn influence subsequent cognition and output.

For example, experience-derived cognition formed at L7 may be developed at L6 into reasons or structures. It may also appear in cognition or output in connection with

verbalization at L1, or with emotional responses at L2. However, the mechanism by which these interactions occur is not addressed in this paper.

Within SCOPE, each layer is treated both as an independent cognitive function and as part of a cognitive structure in which layers mutually influence one another. L7 is no exception, and is positioned as a cognitive layer that functions within interaction with other layers.

This paper has focused primarily on the role of L7 in isolation. The specific interactions and processes involving other layers fall outside the scope of this paper and are intended to be organized in a future paper addressing SCOPE as a whole.

Chapter 5: L7's Relationship with Malkuth

Chapter 4 organized L7 as a cognitive layer that functions within interaction with other layers. This chapter examines the relationship between L7 and Malkuth.

Experience-derived cognition formed at L7 cannot be directly observed from the outside. What can be observed is the output that appears at Malkuth as a result of that cognition — such as speech, text, or behavior.

For example, even when highly abstract, highly compressed expressions such as "it feels a bit blue" or "something's off," or experience-based intuitive judgments, are observed, this alone does not confirm that L7 was at work. Similar output may also arise through interaction with other cognitive layers.

For this reason, SCOPE uses output that appears at Malkuth as a clue to infer the experience-derived cognitive structure behind it. L7 is not a cognitive layer that handles Malkuth itself, but is positioned as one of the internal structures inferred from Malkuth.

This paper has focused primarily on the role of L7 in isolation. The correspondence between the diverse output that appears at Malkuth and L7 falls outside the scope of this paper, and is intended to be organized in a future paper addressing SCOPE as a whole.

Chapter 6: Functional Correspondence with L7 in LLMs

Chapter 5 organized L7 as one of the internal structures inferred from Malkuth. This chapter examines the possibility of a functional correspondence with L7 in LLMs.

LLMs do not undergo the same bodily experience as humans. However, within this paper, experience is defined as an accumulation in which internal structure is updated through learning and subsequently influences later cognition or output. Under this definition, the

learning process in LLMs may also correspond to experience in this sense. As a result, highly abstract, highly compressed expressions or concepts may be reflected in output, and in this respect, a functional correspondence with human L7 may exist.

For example, an LLM acquires abstracted concepts and expressions from large volumes of training data, and can produce them as output depending on context. This functioning differs from the process by which humans form experience, but the function that appears in output may show a correspondence.

This does not indicate that the same cognitive process occurring in humans is actually occurring within an LLM. This paper limits itself to organizing the possibility of correspondence based on observable function.

Final Chapter

This paper has organized L7 as a cognitive layer of highly abstract, highly compressed cognition formed through the accumulation of experience.

L7 does not retain experience in its raw form; rather, it is formed as experience-derived cognition, and may influence subsequent cognition and output. This paper also examined the possibility of a functional correspondence with L7 in LLMs — not only in humans — from the perspective of internal structure updated through learning.

Experience-derived cognition formed at L7 does not exist in isolation. It may relate to other experience-derived cognition, as well as to structures formed at other layers, potentially leading to higher-order integration or overview.

The next paper addresses L8, a cognitive layer involved in the integration and overview of multiple cognitive structures. It will examine how multiple cognitive structures may be integrated and organized as new cognition.